

Lattice-Theoretic Formal Concepts for Sparse Surrogate Feature Analysis in Computer Vision

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Abstract

Sparse surrogate models describe neural activations using a small number of active latent coordinates. We study how these coordinates can be combined into reproducible queries about images and spatial sites. Following a relation-induced construction for language-model features, we derive a lattice-theoretic formal context from nonnegative visual codes. Common descriptions are coordinatewise minima, and retrieval is coordinatewise threshold satisfaction. We retain the Galois-ideal and adjunction proofs and identify the construction with a pattern structure and, on finite data, an ordinally scaled classical context. The vision-specific issue is spatial quantification: max pooling allows different coordinates to be witnessed at different sites. We characterize when one vector can represent same-site evidence and give a downset-valued construction when it cannot. Three digit CNN/autoencoder runs evaluate held-out retrieval and reconstruction. A frozen ResNet34 with a sparse autoencoder supplies measured animal and vehicle feature galleries on a fixed 1,500-image CIFAR-10 subset. Explicit query examples distinguish pooled matches from same-site witnesses. Exhaustive finite checks verify the algebra independently. Released code, weights, images, and activations make the examples auditable; the results do not establish causal or class-exclusive interpretations of the learned coordinates.

Keywords: formal concept analysis; pattern structures; computer vision; sparse autoencoders; spatial feature analysis.

1 Introduction

A vision model can recognize an image accurately while leaving unclear which internal features support its decision. Feature visualization investigates patterns that activate individual units, whereas sparse surrogates learn a larger dictionary in which an activation is represented by relatively few nonzero coefficients [11, 5, 10]. Neither construction makes a coordinate a semantic concept by definition. A useful next question is relational: which images share a collection of features, at which activation levels, and in which spatial locations?

Formal concept analysis (FCA) organizes objects and their shared attributes into a concept lattice [4]. In the present setting, the objects are images or spatial sites, and an attribute query specifies minimum values for selected sparse coordinates. For example, a query requiring two features at levels 2 and 1 retrieves precisely the objects whose codes dominate $(2, 1)$. The common description of those objects is their coordinatewise minimum. Closing the query replaces it by this common description without changing its retrieved objects. This is a mathematical account of shared evidence, not a guarantee that the evidence has the interpretation suggested by a feature visualization.

The paper adapts the summary–retrieval framework of the companion language-model manuscript [17]. We preserve its sequence of arguments: construct a Galois ideal, derive the antitone Galois connection, identify the context derivations, and relate the resulting concepts to classical FCA. These arguments are applications of established theory [19, 3]; we make no novelty claim for the existence of the adjunction or concept lattice.

Images introduce a distinction that cannot be handled by changing the word “token” to “patch.” If one patch exhibits the first feature and another exhibits the second, their coordinatewise maximum satisfies the two-feature query even though neither patch satisfies it. The difference is between “for each coordinate, some site” and “some site, for every coordinate.” We formalize both readings, characterize the special case in which a single vector represents them equally, and use a lattice of downsets to retain same-site evidence in general. Site-level evidence itself need not be spatially local in the input: attention and convolution can give a site a receptive field larger than its displayed cell.

Our empirical contribution combines a controlled digit pilot with a natural-image feature study. The digit pilot uses a convolutional classifier, a nonnegative Top- k sparse autoencoder, separate training, calibration, and test images, and saved query vectors and scores. A frozen pretrained ResNet34 supplies animal and vehicle examples. Public vision libraries supply a route to further models; their published benchmark numbers are not results of this paper. Our studies test whether the proposed operations can be computed and interpreted consistently. They do not test foundation-model superiority or establish a new state of the art.

The exposition first specifies sparse surrogates and spatial summaries, then develops the context and its spatial refinement. We subsequently report computational checks, the digit benchmark, and natural-image examples, and conclude with the scope of the evidence and requirements for a larger evaluation.

2 Sparse Visual Features and Their Summaries

The construction needs nonnegative feature descriptions and a declared object set. It does not depend on a particular neural architecture. We separate the feature extractor, the sparse surrogate, and the spatial summary so that each operation has an explicit interpretation.

2.1 Activation states and sparse surrogates

Let X be a finite nonempty image collection. For image $x \in X$, let P_x be a finite nonempty set of spatial sites. A frozen vision model provides an activation $h_{x,p} \in \mathbb{R}^m$ at site $p \in P_x$ of a fixed layer. In a convolutional network these sites index a feature map; in a vision transformer they can index patch tokens. Classification and register tokens must be handled separately if the intended interpretation is spatial.

A sparse autoencoder (SAE) encodes and reconstructs the same activation:

$$z_{x,p} = E(h_{x,p}) \in \mathbb{R}_{\geq 0}^d, \quad \hat{h}_{x,p} = D(z_{x,p}), \quad D(z) = W_{\text{dec}}z + b_{\text{dec}}. \quad (1)$$

Typically $d > m$. A nonnegative Top- k encoder takes a rectified linear projection and retains its k largest entries, so its code has at most k nonzero coordinates [5]. With this hard sparsity constraint, one can minimize squared reconstruction error. An alternative is a rectified encoder with an ℓ_1 penalty:

$$\mathcal{L}_{\text{SAE}} = \mathbb{E}_h [\|D(E(h)) - h\|_2^2 + \lambda \|E(h)\|_1]. \quad (2)$$

The pilot uses the hard Top- k constraint, not the penalized objective. Decoder normalization removes a simple source of arbitrary scaling; training and random initialization can still change the dictionary.

A transcoder instead takes the input $a_{x,p}$ to a selected transformation and predicts its output $h_{x,p}$:

$$z_{x,p} = E_{\text{TC}}(a_{x,p}), \quad \hat{h}_{x,p} = D_{\text{TC}}(z_{x,p}). \quad (3)$$

It is a sparse surrogate for a computation rather than merely for an activation state [1]. Both models fit the theory when their codes are nonnegative. Their coordinates are not directly comparable across dictionaries, even when dimensions coincide. The pilot below evaluates SAEs only; no SAE-transcoder advantage is inferred.

2.2 Images, regions, and spatial sites

For an image-level analysis, define its summary by

$$u_{x,j} = \max_{p \in P_x} z_{x,p,j}, \quad 1 \leq j \leq d. \quad (4)$$

This is an existential summary for each coordinate separately. It retains activation magnitudes but discards which site supplies each maximum. A region-level analysis can instead use objects (x, S) with nonempty $S \subseteq P_x$ and take the maximum over S . A site-level analysis uses objects (x, p) and assigns the unpooled code $z_{x,p}$. All three choices yield the same kind of graded context; they answer different questions.

For a fixed image context, let

$$M_j = \max_{x \in X} u_{x,j}, \quad \mathcal{U} = \prod_{j=1}^d [0, M_j], \quad \mathcal{X} = \mathcal{P}(X). \quad (5)$$

Order $\mathcal{U}$ coordinatewise and $\mathcal{X}$ by inclusion. A complete lattice is an ordered set in which every family, including the empty family, has a least upper bound (join) and a greatest lower bound (meet). In $\mathcal{U}$ these operations are coordinatewise suprema and infima, with bottom 0 and top M . In $\mathcal{X}$ they are unions and intersections, with bottom $\emptyset$ and top X . A coordinate with $M_j = 0$ simply contributes a one-element factor. Sparsity of site codes need not survive max pooling: different sites can activate different coordinates.

The bound M is specific to the context on which closure is computed. For held-out retrieval, a frozen training query is compared directly with new codes without clipping them at M . Thus the experiment does not silently use test activations to choose training thresholds. A larger context can be defined afterwards, but its closure operator can differ.

2.3 Available vision libraries

Public implementations make the framework applicable beyond the pilot. ViT-Prisma supplies activation access, vision SAEs, and CLIP transcoders; its published resources include CLIP and DINO dictionaries [8, 13]. SAEV provides a vision SAE training and inference workflow with linked Hugging Face checkpoints [7, 16]. Overcomplete offers several SAE and dictionary-learning variants together with visualization and evaluation tools [2]. PatchSAE provides closely related evidence that sparse visual codes can support patch-level analysis during model adaptation [10].

For a foundation-model extension, ViT-Prisma is the first candidate when both SAEs and transcoders are required; SAEV is a natural alternative for vision SAE experiments, and Overcomplete for controlled training studies. This is a choice of software scope, not a measured performance ranking. A checkpoint is usable only with its matching backbone, layer, activation normalization, preprocessing, and token selection. The implementation therefore accepts cached nonnegative site codes as its core interface. The executed pilot uses the project’s existing PyTorch environment and a small explicitly defined surrogate, rather than claiming to have tested these external libraries.

3 The Relation-Induced Context

We now give the mathematical construction independently of the neural model. This preserves the proof pattern of the companion paper and makes explicit which properties come from order theory rather than from training. Throughout this section, $\mathcal{U}$ can be any complete lattice.

3.1 From individual summaries to a Galois ideal

Let X be a set and $J \subseteq X \times \mathcal{U}$ any relation. Assign each object the summary

$$u_x = \bigwedge \{u \in \mathcal{U} \mid (x, u) \in J\}. \quad (6)$$

The empty meet is $\top_{\mathcal{U}}$. For visual data, J is the graph of the chosen summary map, so this meet is over a singleton; it does not replace the spatial maximum in Equation (4). Put $\mathcal{X} = \mathcal{P}(X)$ and

$$(A, u) \in I \iff u \leq u_x \text{ for every } x \in A. \quad (7)$$

Here $A, B \in \mathcal{X}$ denote collections of objects and $u, v \in \mathcal{U}$ denote attribute descriptions. Individual images are indexed by x .

Definition 3.1 (Galois ideal and lattice-theoretic context). For complete lattices $\mathcal{X}$ and $\mathcal{U}$, a relation $I \subseteq \mathcal{X} \times \mathcal{U}$ is a Galois ideal if it is a lower set in the product order and is closed under arbitrary joins in each coordinate separately. Thus, including empty families,

$$\begin{aligned} (A_t, u) \in I \ (t \in T) &\implies (\bigvee_{t \in T} A_t, u) \in I, \\ (A, u_t) \in I \ (t \in T) &\implies (A, \bigvee_{t \in T} u_t) \in I. \end{aligned} \quad (8)$$

The triple $(\mathcal{X}, \mathcal{U}, I)$ is then a lattice-theoretic formal context [19].

Proposition 3.2 (Relation-induced Galois ideal). *The relation in Equation (7) is a Galois ideal.*

Proof. Suppose $(A, u) \in I$, $B \subseteq A$, and $v \leq u$. For every $x \in B$, we have $v \leq u \leq u_x$, so $(B, v) \in I$. This proves the lower-set property. If $(A, u_t) \in I$ for every $t \in T$, each u_x with $x \in A$ is an upper bound of the family, hence $\bigvee_{t \in T} u_t \leq u_x$. If $(A_t, u) \in I$ for every $t \in T$, every $x \in \bigcup_{t \in T} A_t$ belongs to some A_t and satisfies $u \leq u_x$. The first assertion also holds for an empty family because its join is $\perp_{\mathcal{U}}$; the second is then vacuous. $\square$

3.2 Derivations, closure, and concepts

The summary and retrieval maps are

$$F_I(A) = \bigwedge_{x \in A} u_x, \quad G_I(u) = \{x \in X \mid u \leq u_x\}. \quad (9)$$

Thus $F_I(\emptyset) = \top_{\mathcal{U}}$ and $G_I(\perp_{\mathcal{U}}) = X$.

Proposition 3.3 (Summary–retrieval adjunction). *The maps $F_I : \mathcal{X} \rightarrow \mathcal{U}$ and $G_I : \mathcal{U} \rightarrow \mathcal{X}$ are antitone and satisfy*

$$A \subseteq G_I(u) \iff u \leq F_I(A). \quad (10)$$

Consequently, $c_I = G_I F_I$ and $d_I = F_I G_I$ are closure operators on $\mathcal{X}$ and $\mathcal{U}$, respectively.

Proof. If $B \subseteq A$, taking a meet over the larger set gives $F_I(A) \leq F_I(B)$. If $u \leq v$, every summary dominating v also dominates u , so $G_I(v) \subseteq G_I(u)$. Finally, $A \subseteq G_I(u)$ means exactly that $u \leq u_x$ for all $x \in A$, equivalently $u \leq \bigwedge_{x \in A} u_x$.

The equivalence gives $A \subseteq G_I F_I(A)$ and $u \leq F_I G_I(u)$. The composites are monotone since each factor is antitone. Applying antitonicity to these inequalities gives $F_I G_I F_I = F_I$ and $G_I F_I G_I = G_I$, hence idempotence of both composites. These are precisely extensivity, monotonicity, and idempotence, the defining properties of closure operators. $\square$

Proposition 3.4 (Identification of context derivations). *The lattice-theoretic derivations associated with I are exactly $A^I = F_I(A)$ and $u^I = G_I(u)$.*

Proof. By definition, u^I is the join of all A incident with u , here $\bigcup \{A \in \mathcal{X} \mid (A, u) \in I\}$. Every such A is contained in $G_I(u)$; conversely, each $x \in G_I(u)$ supplies the incident set $\{x\}$. Thus this union is $G_I(u)$. Similarly, A^I is the join of all u incident with A . Every such u lies below $F_I(A)$, and $F_I(A)$ is itself incident with A , proving equality. $\square$

A lattice-theoretic formal concept is a pair (A, u) satisfying $F_I(A) = u$ and $G_I(u) = A$. The set A is its *extent*, and u is its *intent*. Order the concepts by

$$(A, u) \leq (B, v) \iff A \subseteq B \iff v \leq u. \quad (11)$$

A larger extent has a weaker shared description. Closed concepts record all consequences of a query on the chosen dataset, not on unseen images.

Theorem 3.5 (Concept lattice and generated concepts). *The concepts form a complete lattice $\mathfrak{B}_I$. For any indexed family $\{(A_t, u_t)\}_{t \in T}$,*

$$\begin{aligned} \bigwedge_{t \in T} (A_t, u_t) &= \left(\bigcap_{t \in T} A_t, d_I \left(\bigvee_{t \in T} u_t \right) \right), \\ \bigvee_{t \in T} (A_t, u_t) &= \left(c_I \left(\bigcup_{t \in T} A_t \right), \bigwedge_{t \in T} u_t \right). \end{aligned} \quad (12)$$

The generated concepts are

$$\gamma(A) = (c_I(A), F_I(A)), \quad \mu(u) = (G_I(u), d_I(u)). \quad (13)$$

Proof. Closed sets of a closure operator are closed under arbitrary intersections: if each A_t is closed, monotonicity gives $c_I(\bigcap_t A_t) \subseteq A_t$ for every t , and extensivity gives the reverse inclusion to their intersection. Their least closed upper bound is the closure of their union. Also, directly from the adjunction,

$$G_I \left(\bigvee_{t \in T} u_t \right) = \bigcap_{t \in T} G_I(u_t), \quad F_I \left(\bigcup_{t \in T} A_t \right) = \bigwedge_{t \in T} F_I(A_t). \quad (14)$$

Together with $F_I G_I F_I = F_I$, these identities give the stated intents and establish both formulas, including empty families. The same identities prove that $\gamma(A)$ and $\mu(u)$ are concepts. $\square$

The maps in Equation (13) need not be embeddings: distinct queries or source sets can generate the same concept. In particular, $G_I(d_I(u)) = G_I(u)$. A join of raw queries conjoins requirements, whereas a meet relaxes them:

$$G_I(u \vee v) = G_I(u) \cap G_I(v), \quad G_I(u \wedge v) \supseteq G_I(u) \cup G_I(v). \quad (15)$$

The inclusion can be strict. These operations on queries should not be confused with the extent-ordered concept operations in Equation (12).

3.3 A worked example and classical representation

Suppose three images have summaries

$$u_{x_1} = (4, 1), \quad u_{x_2} = (2, 3), \quad u_{x_3} = (1, 0). \quad (16)$$

Then $\mathcal{U} = [0, 4] \times [0, 3]$. The query $(2, 1)$ retrieves x_1, x_2 and is already closed. The query $(3, 1)$ retrieves only x_1 and closes to $(4, 1)$. Both queries have the same positive support, so a presence-only representation cannot distinguish them. In the other direction, the singleton $\{x_3\}$ closes to all three images because every summary dominates $(1, 0)$. Finally,

$$\begin{aligned} G_I((3, 0) \vee (0, 2)) &= \emptyset, \\ G_I((3, 0) \wedge (0, 2)) &= X \not\supseteq \{x_1, x_2\}. \end{aligned} \quad (17)$$

Here a stronger conjunction is empty, whereas the shared lower bound is a completely unselective query. Neither outcome is mathematically exceptional.

For a functional summary map, this construction is a pattern structure: each object has a description in a meet-semilattice, and common descriptions are obtained by meets [3]. It is also representable by classical FCA without discarding magnitudes.

Proposition 3.6 (Observed-threshold representation). *Let X be finite and nonempty, with nonnegative real summaries and $\mathcal{U}$ as in Equation (5). Define classical attributes*

$$E = \{(j, a) \mid 1 \leq j \leq d, a > 0, a = u_{x,j} \text{ for some } x \in X\}. \quad (18)$$

Declare x incident with (j, a) exactly when $u_{x,j} \geq a$. This classical context and the vector context have identical closed extents; their concept lattices are therefore isomorphic by preservation of extents.

Proof. The classical intent of $A \subseteq X$ consists precisely of the pairs (j, a) with $a \leq F_I(A)_j$. Each coordinate of $F_I(A)$ is an observed value or zero, including $A = \emptyset$, whose meet is M . Thus satisfying every attribute in that classical intent is equivalent to dominating $F_I(A)$. Consequently $A'' = G_I F_I(A)$. Since a concept is determined by its extent and ordered by extent inclusion, the equality of extent closures proves the isomorphism. $\square$

An arbitrary $u \in \mathcal{U}$ can be rounded coordinatewise upward to the next observed value without changing its extent. Presence-only FCA retains just one bit per coordinate, whereas ordinal scaling retains the relevant thresholds. Any advantage over a presence baseline is therefore an advantage of retaining amplitudes, not an expressive advantage over FCA. The vector representation gives these amplitudes and their operations directly, without materializing a larger binary incidence matrix.

4 Spatial Witnesses and the Limits of Pooling

The preceding construction is valid for any chosen summaries. Its interpretation nevertheless depends on how they were formed. This section isolates the spatial issue and gives an exact representation when max pooling is too coarse.

4.1 Separate-site and same-site evidence

For a query $u \in \mathcal{U}$, define two image retrieval sets:

$$\begin{aligned} G_{\text{img}}(u) &= \{x \mid \forall j \exists p \in P_x : u_j \leq z_{x,p,j}\}, \\ H(u) &= \{x \mid \exists p \in P_x \forall j : u_j \leq z_{x,p,j}\}. \end{aligned} \tag{19}$$

The first is $G_I(u)$ for max-pooled summaries. The second requires a single site to witness every coordinate. It is also the image projection of the extent in the site-level context.

Proposition 4.1 (Spatial relaxation under max pooling). *For every query, $H(u) \subseteq G_{\text{img}}(u)$. Equality holds for queries with at most one positive coordinate, but can fail for two positive coordinates. In general, H need not take joins to intersections.*

Proof. A site witnessing every coordinate supplies a witness for each coordinate separately. With one positive coordinate, a site attaining its maximum witnesses that coordinate, while nonnegativity satisfies the others. For a zero query, every image qualifies since each P_x is nonempty. For strictness, take one image with codes $(2, 0)$ and $(0, 2)$. It belongs to $G_{\text{img}}((2, 2))$ but not to $H((2, 2))$. It also belongs to both $H((2, 0))$ and $H((0, 2))$, proving failure of join-to-intersection preservation. $\square$

The example prevents treating H as the retrieval half of the same vector-lattice adjunction: any antitone Galois retrieval map must satisfy Equation (14). It also suggests an observable diagnostic. When $G_{\text{img}}(u) \neq \emptyset$, define the witness discrepancy

$$\Delta(u) = 1 - \frac{|H(u)|}{|G_{\text{img}}(u)|}. \tag{20}$$

This measures the fraction of pooled matches lacking a same-site witness. It is undefined for an empty pooled extent and is not a measure of semantic error: dispersed evidence may be exactly what a query intends.

4.2 When a vector suffices, and when a downset is needed

Write $\downarrow v = \{u \in \mathcal{U} \mid u \leq v\}$. The set of queries witnessed at some site of image x is

$$s_x = \bigcup_{p \in P_x} \downarrow z_{x,p}. \tag{21}$$

It is a downset: whenever it contains a query, it contains all weaker queries. Max pooling replaces it by the generally larger principal downset $\downarrow u_x$.

Proposition 4.2 (Single-vector criterion for spatial evidence). *For a finite nonempty P_x , the downset s_x is principal if and only if one site code dominates all the other site codes. In that case it equals $\downarrow u_x$.*

Proof. If one site code dominates all the others, it is their coordinatewise maximum, and the union in Equation (21) is its principal downset. Conversely, suppose $s_x = \downarrow v$. Every site code lies below v , so $u_x \leq v$. Because $v \in s_x$, some site p satisfies $v \leq z_{x,p} \leq u_x$. Hence $v = z_{x,p} = u_x$, and that site code dominates all the others. $\square$

The criterion describes a restrictive special case, not a generic property of sparse representations. To retain same-site evidence, let $\mathcal{D}(\mathcal{U})$ be the set of all downsets of $\mathcal{U}$, including the empty set, ordered by inclusion. Arbitrary unions and intersections are its joins and meets; it is a complete lattice with bottom $\emptyset$ and top $\mathcal{U}$. Use s_x as the description of image x and define

$$F_{\text{sp}}(A) = \bigcap_{x \in A} s_x, \quad G_{\text{sp}}(u) = \{x \mid u \subseteq s_x\}, \quad u \in \mathcal{D}(\mathcal{U}). \quad (22)$$

Here u denotes a downset-valued attribute, rather than a vector.

Theorem 4.3 (Downset context for spatial evidence). *The maps in Equation (22) form an antitone Galois connection. For every vector $v \in \mathcal{U}$,*

$$G_{\text{sp}}(\downarrow v) = H(v). \quad (23)$$

Thus same-site vector queries admit exact concept closure after passing to downset descriptions.

Proof. The equivalence $A \subseteq G_{\text{sp}}(u)$ if and only if $u \subseteq \bigcap_{x \in A} s_x$ proves the adjunction, just as in Proposition 3.3. Since s_x is downward closed, $\downarrow v \subseteq s_x$ is equivalent to $v \in s_x$, which is exactly the existence of a site code dominating v . $\square$

This uses the established pattern-structure construction with richer descriptions. It does not make same-site retrieval preserve vector joins: $\downarrow v \cup \downarrow w$ generally differs from $\downarrow(v \vee w)$. The former can be witnessed at different sites, whereas the latter requires a site satisfying the joint vector. Closure may likewise turn a principal downset into a nonprincipal one.

The representation is computationally concrete. A finite union of principal downsets is determined by its maximal generating vectors. Intersections can be computed using

$$\left(\bigcup_i \downarrow v_i \right) \cap \left(\bigcup_j \downarrow w_j \right) = \bigcup_{i,j} \downarrow (v_i \wedge w_j). \quad (24)$$

Dominated generators can then be discarded. The number of undominated generators can grow rapidly, so the pilot computes H by direct site comparisons and does not enumerate the downset concept lattice. This preserves the distinction between an exact representation theorem and a scalable mining algorithm.

5 Computational Verification and an Image Pilot

We separate exact checks of the algebra from an experiment on learned visual features. The former can detect implementation errors; the latter measures reconstruction and class-associated retrieval in one small setting. Neither substitutes for the other.

5.1 Exact finite checks

The executable audit enumerates all $3^6 = 729$ matrices with three objects, two coordinates, and entries in $\{0, 1, 2\}$. For each matrix it checks all eight object subsets and every integer query in the matrix-bounded product lattice. This gives 41,472 checks of Equation (10). It additionally checks object-closure extensivity and idempotence, preservation of query extents under closure, join-to-intersection retrieval, and equality with an independently constructed ordinally scaled binary closure. Empty extents, zero coordinates, and all-zero matrices are included.

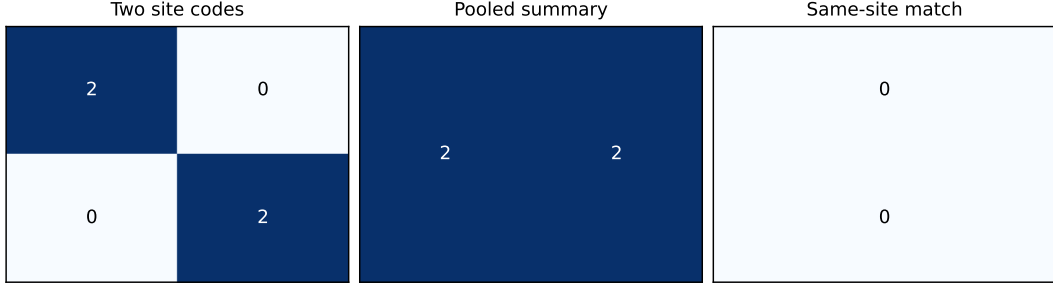


Figure 1: Constructed example for query (2, 2). The pooled summary satisfies both coordinates, but neither site does. Matrix rows in the first panel are sites and columns are features. The last panel records the two failed same-site tests. These are illustrative codes, not activations measured from an image.

Table 1: Held-out classifier and surrogate diagnostics.

Seed	Accuracy	Replaced	R_{tr}^2	Site nnz	Image nnz	Alive
17	96.39	95.00	0.889	8.00	15.67	58
29	97.22	95.00	0.905	8.00	16.66	53
43	96.39	95.56	0.925	8.00	14.95	49

A separate enumeration covers all $3^4 = 81$ two-site, two-coordinate code arrays and nine queries per array, giving 729 checks of the spatial inclusion in Proposition 4.1. The counterexample with site codes (2, 0) and (0, 2) is tested explicitly. All these checks pass. They verify finite instances and implementation conventions, not the universal theorems, whose proofs were given above.

5.2 Data, training, and reconstruction

We use the 1,797 grayscale 8×8 images supplied by scikit-learn’s `load_digits` [15]. Pixel values are divided by 16. Two stratified splits with random state 2026 yield 1,078 training, 359 calibration, and 360 test images. The split is by image, not by writer; writer-independent generalization is not tested. The dataset is the small bundled digits collection, not MNIST or a natural-image corpus.

Each classifier has a padded 3×3 convolution from 1 to 16 channels, ReLU, 2×2 max pooling, a padded 3×3 convolution from 16 to 32 channels, ReLU, and a linear ten-class head on the flattened $4 \times 4 \times 32$ feature map. We train for 25 epochs with Adam, learning rate 0.001, batch size 64, and cross-entropy loss. The analysis sites are the 16 cells after the second ReLU. Their overlapping receptive fields matter: “same site” refers to this feature map, not necessarily to one small isolated part of the original image.

For each frozen classifier, we train a 32-to-128-to-32 SAE with a linear encoder, ReLU, and Top-8 selection, using only training-image sites. The decoder is linear with a bias and unit-length columns, normalized after every update. Training uses 20 epochs of Adam at learning rate 0.001, batch size 512, and mean squared error, without centering or an additional sparsity penalty. Classifier and SAE training are repeated with seeds 17, 29, and 43; all other choices, including the split, remain fixed. The implementation runs deterministically on CPU with two threads. No checkpoint or hyperparameter is selected using test performance.

In Table 1, “Accuracy” and “Replaced” are test classification percentages before and after

replacing all analyzed site activations with SAE reconstructions. The reconstruction statistic is

$$R_{\text{tr}}^2 = 1 - \frac{\sum_{x \in X_{\text{te}}} \sum_{p \in P_x} \|\hat{h}_{x,p} - h_{x,p}\|_2^2}{\sum_{x \in X_{\text{te}}} \sum_{p \in P_x} \|h_{x,p} - \bar{h}_{\text{tr}}\|_2^2}, \quad (25)$$

where $\bar{h}_{\text{tr}}$ is the mean over training sites. It uses a training mean rather than fitting a mean on the test set. “Site nnz” and “Image nnz” are the mean numbers of positive coordinates before and after spatial max pooling. “Alive” counts coordinates active anywhere in the training codes.

Reconstruction retains much, but not all, of the classifier’s performance. Only 49–58 of the 128 coordinates are active on training data, so this pilot is not evidence of an optimized dictionary. Pooling increases the mean active-coordinate count from 8 to approximately 15–17. This is a concrete reason to report the object granularity along with sparsity.

5.3 Queries, calibration, and baselines

Each digit supplies a one-versus-rest retrieval task. For each model seed and digit, we draw five sets of three positive training images without replacement within a draw; draws can overlap. There are 50 queries per model and 150 in total. Let A_0 denote one source set and put $v = F_I(A_0)$, using max-pooled training codes. Rank coordinates by $v_j / \max(s_j, 10^{-12})$, where s_j is the root mean square of training-image coordinate j . Ties are broken by coordinate index. For the first b ranked coordinates, denoted T_b , define

$$u_j = \begin{cases} \alpha v_j, & j \in T_b, \\ 0, & j \notin T_b. \end{cases} \quad b \in \{1, 4, 16, 128\}, \quad \alpha \in \{0.25, 0.5, 0.75, 1\}. \quad (26)$$

A coordinate budget can exceed the number of positive coordinates of v ; it is an upper bound on query support. This query construction is a heuristic for selecting sparse evidence, not a lattice-theoretic optimum.

We choose (b, α) separately for each query by maximizing calibration F_1 , the harmonic mean of precision and recall, with zero returned when its denominator is zero. Ties follow the displayed increasing budget and multiplier order. The classifier, SAE, coordinate scales, and positive source examples all come from training data; calibration labels select the query parameters. Test images are used only afterwards.

For ranking, the graded satisfaction score is

$$\rho_u(x) = \min_{j: u_j > 0} \frac{u_{x,j}}{u_j}. \quad (27)$$

For a zero query we use the constant score 1. Thus $\rho_u(x) \geq 1$ is exactly pooled query satisfaction. We report average precision (AP), summing precision over recall increments at distinct score thresholds. This is a ranking metric; the fixed-cutoff F_1 measures a different aspect of retrieval. Scaling all positive query coordinates by a common α leaves their ranking unchanged, but changes thresholded predictions and can affect the selected budget.

The presence baseline uses the same training-based coordinate ranking and candidate budgets, replaces each selected positive requirement by positivity, and selects its budget by calibration F_1 . Its ranking score is the fraction of these requirements met; its extent requires all of them. An empty requirement set receives constant score 1. This is an *amplitude-informed selection with presence-only scoring*, not a fully binary learning pipeline. It isolates a specific loss of magnitude information; it does not cover every FCA baseline.

Table 2: Mean test average precision (percent), 50 queries per seed.

Method	17	29	43	Mean
Graded conjunction	35.65	31.80	36.37	34.60
Presence baseline	11.83	12.47	13.29	12.53
Sparse-code cosine	53.96	54.70	51.48	53.38
Dense-code cosine	66.44	67.03	68.10	67.19
Random sources	12.83	13.12	13.02	12.99

Table 3: Thresholded retrieval and spatial witnesses on test images.

Seed	Graded F_1	Binary F_1	Pooled	Same-site	Mean Δ
17	0.359	0.211	4371	1347	0.667
29	0.333	0.224	4877	1776	0.610
43	0.362	0.232	4000	1232	0.720

Two cosine baselines compare each test summary with the mean summary of the same three source images, using either max-pooled SAE codes or the 32-dimensional dense site activations. Zero-norm comparisons receive score zero. A random-source diagnostic replaces the positive examples by three arbitrary training images, using the selected graded budget and multiplier without recalibration. It tests source dependence rather than serving as an independently optimized competitor. All methods see the same held-out images. The comparison concerns retrieval from pooled representations, not the classifier’s spatially ordered linear head.

5.4 Results and their interpretation

The graded query improves mean AP over the specified presence baseline for all three seeds, but both cosine baselines perform better (Table 2). Dense-code cosine gives mean AP 67.19%, compared with 34.60% for graded queries and 12.53% for presence scoring. The pilot therefore supports a limited conclusion: amplitudes help within this conjunctive retrieval design, while these sparse conjunctions are not the strongest class-retrieval method among those tested.

The test F_1 scores in Table 3 likewise favor graded over presence-only satisfaction. Every calibrated graded query has a nonempty test extent. Budgets 1, 4, and 16 are selected 49, 59, and 42 times, respectively; budget 128 is never the first maximizing choice. This does not show that all 128-coordinate candidates perform strictly worse: with few shared positive coordinates they can coincide with smaller budgets and lose the specified tie break.

The pooled and same-site totals count query–image matches, so one image can contribute to many queries. The last column averages Equation (20) over the 50 queries for each seed. Its values show that roughly 61–72% of pooled matches per query lack a same-site witness on average. This includes zero discrepancy for queries with one positive coordinate. The discrepancy is a property of the chosen spatial quantifiers; it should not be interpreted as an error rate for the model.

Closure preserves the training extent for all 150 chosen queries, as required by the theory. We do not replace them by training-closed queries when evaluating new images: the equality of extents under closure is relative to the training context and need not survive a change of object set. The stored queries and scores make this choice explicit.

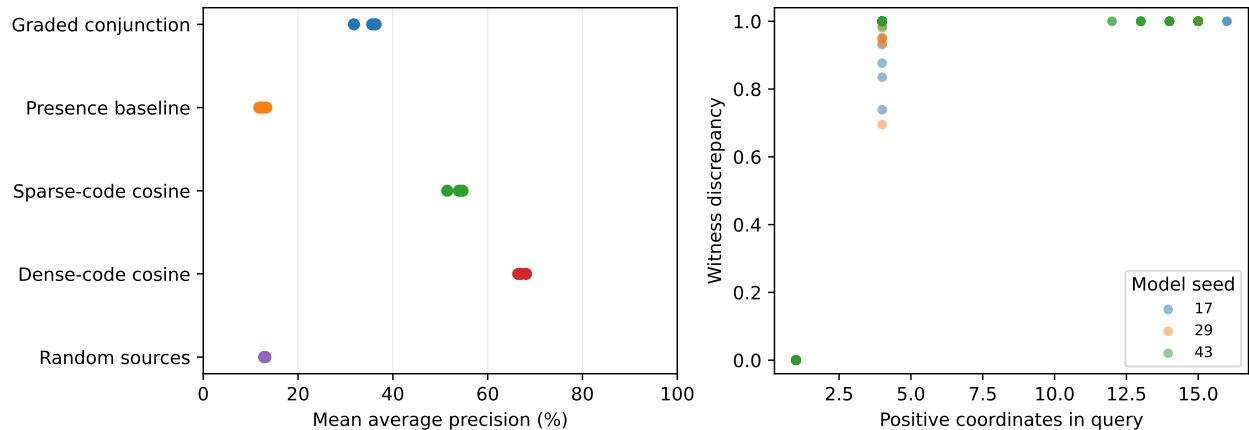


Figure 2: Left: mean test AP for each of the three model seeds. Right: witness discrepancy versus actual query support for all 150 graded queries. Points overlap. These plots describe one fixed image split; the seed variation is not a confidence interval over datasets.

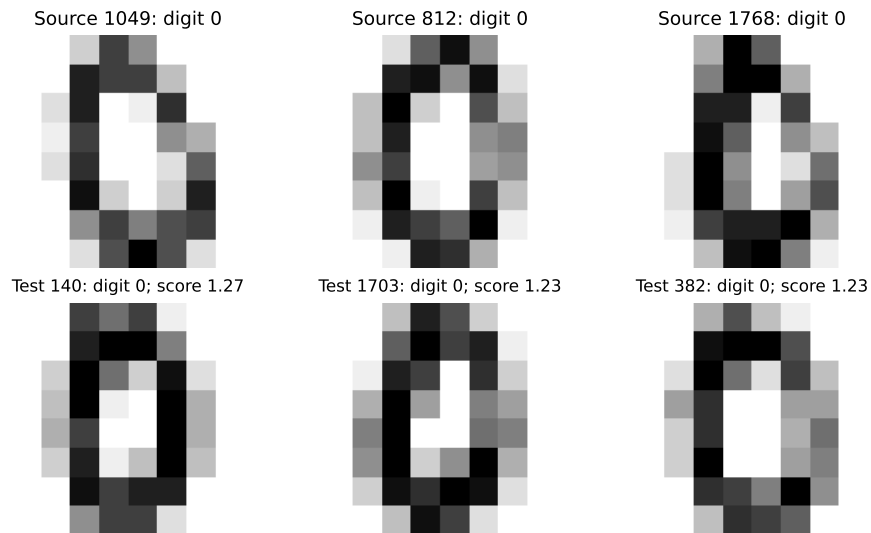


Figure 3: An inspectable retrieval example, selected as the first query in the fixed protocol: seed 17, digit 0, source draw 0. Top: its three training images. Bottom: the three highest-scoring test images under the graded query, with ground-truth digit labels and scores from Equation (27). This example is not a semantic annotation of the selected latent coordinates.

5.5 Reproducibility and computational scope

The source and reproducibility artifacts are available in the public Lattice-ML repository [18]: <https://github.com/levants/lattice-ml>. The package separates the context operations, neural models, experiment runner, algebraic audit, and table generator. The companion Jupyter notebook reloads cached evidence and checks the reported results. Saved artifacts include model and SAE weights, site codes, dense activations, source-image indices, all split indices, query vectors, test scores, package versions, and SHA-256 hashes. Tables and figures are generated from the results file rather than edited independently. The code uses the existing project environment through `uv`. Both repositories use the same relative layout: Python modules in `src/lattmc/vision`, notebooks in `notebooks/vision`, and data in `vision_tokens`. Within each experiment, datasets, checkpoints, activation caches, retrieval arrays, and result summaries are stored separately. Dataset indices and content hashes identify the exact images and arrays underlying every example.

For n image summaries and a query with q positive coordinates, retrieval costs $O(nq)$ comparisons. Computing its full closed intent costs $O(nd)$ in the worst case. Direct same-site testing costs $O(nPq)$ when each image has P sites. The implementation stores dense arrays for this small pilot; sparse formats or batched scans are needed at larger scale. We do not enumerate the full concept lattice, whose number of concepts can be exponential in the number of objects.

6 Visual Feature Examples on Natural Images

The digit benchmark establishes an executable protocol, but it provides only a narrow view of visual features. We now examine a pretrained image encoder on animals and vehicles. Following the investigative approach of *Zoom In* [12], we combine highly activating dataset examples, spatial response maps, and explicit comparisons between feature requirements. These are newly computed figures, not reproductions of Distill’s images or evidence for its InceptionV1 circuits.

6.1 A fixed encoder and an inspectable sample

We use ResNet34 [6] with torchvision’s `IMAGENET1K_V1` weights [14] and CIFAR-10 [9]. For each of the ten CIFAR-10 classes, take the first 125 images in the official training order and the first 25 in the official test order. The first 100 selected training images per class train the surrogate; the remaining 25 form a reserved calibration set. Thus there are 1,000 training, 250 calibration, and 250 test images. The examples below use fixed rules and do not tune on the reserved calibration set. This deterministic subset is illustrative, not a representative random sample of all natural images.

Each 32×32 RGB image is resized bilinearly to 96×96 and normalized using the ImageNet channel means and standard deviations. The smaller resolution is a declared departure from the weight package’s standard evaluation transform, chosen to keep extraction inexpensive. We cache the output of the complete third residual stage, after its last addition and ReLU: a 6×6 grid of 256-dimensional activations. The backbone remains frozen and its ImageNet classifier is not evaluated as a CIFAR-10 classifier.

Each channel is divided by its root mean square on training sites, floored at 10^{-6} . A 256-to-512-to-256 Top-16 SAE is trained on these scaled activations for 12 epochs with Adam, learning rate 0.001, batch size 512, and seed 2026. Decoder columns are normalized after each update. The reconstruction is rescaled to the original channel units before computing Equation (25). Test reconstruction gives $R_{\text{tr}}^2 = 0.576$, and 165 of the 512 latent coordinates are active on training data. This moderate fidelity and substantial inactive fraction limit the interpretability claim; the experiment is not a comparison of optimized surrogate architectures.

6.2 Feature galleries and response profiles

For a class c , select a coordinate by its training contrast

$$a_{c,j} = \frac{\text{mean}_{x \in X_{\text{tr}}, y_x=c} u_{x,j} - \text{mean}_{x \in X_{\text{tr}}, y_x \neq c} u_{x,j}}{\max(\text{std}_{x \in X_{\text{tr}}} u_{x,j}, 10^{-12})}. \quad (28)$$

Here y_x is the dataset class and the standard deviation uses divisor $|X_{\text{tr}}|$. Process bird, cat, dog, horse, ship, and truck in that order, taking the highest-contrast coordinate not selected previously; break ties by coordinate index. This selects coordinates 271, 425, 150, 358, 78, and 461. The class names describe the selection criterion, not established meanings of the coordinates.

For each selected coordinate, Figures 4 and 5 show its three highest-activating test images across *all* classes, paired with the unpooled 6×6 code maps. A common color range within each row permits comparison between its examples: black denotes zero, and brighter colors denote larger activations. Labels and original CIFAR test indices make mismatches inspectable. The low-resolution images are displayed without invented detail. Site maps are response locations in the feature grid, not segmentation masks or causal attribution maps: the residual network’s receptive fields can cover most or all of an image.

Figure 6 summarizes the entire 250-image test sample. Each cell is the mean max-pooled activation for one class, divided by the coordinate’s training-image root mean square. It is a class response profile, not a controlled stimulus-tuning experiment. The single-feature test AP values for the six selected classes are respectively 0.176, 0.338, 0.456, 0.366, 0.451, and 0.514. Equal class prevalence is 0.1. These descriptive associations neither imply class exclusivity nor show that a coordinate causes the backbone’s prediction.

6.3 Animal and vehicle queries under meet and join

To parallel the companion paper’s comparisons of related examples, fix two pairs of associated classes: cat/dog and ship/truck. For each class, choose the training image with maximal activation of its selected coordinate, breaking ties by the stored sample order. Keep the two coordinates selected for that pair. The source query retains half of each corresponding source-image activation and is zero elsewhere. Thus both queries use the same coordinate set, even when a source is selected for only one of its coordinates. The multiplier 0.5 is fixed, not selected using test performance. Displayed thresholds are rounded; counts use the full-precision values saved in the retrieval arrays.

Figure 7 shows the cat/dog queries on coordinates 425 and 150. Their test extents contain 40 and 25 images; the meet retrieves 74 and the join 17. The join extent is exactly the intersection of the two source extents. All four same-site extents are empty. Hence these apparent combinations of animal-associated evidence rely on separate spatial witnesses at the chosen thresholds.

For ship/truck-associated coordinates 78 and 461, the source extents contain 10 and 23 images, the meet contains 158, and the join contains one automobile image. Their respective same-site counts are 2, 7, 72, and 0. The sole join match is therefore not an image labeled both ship and truck, and it has no common spatial witness. The algebra combines activation requirements; its result is not a conjunction of class names.

These examples make the semantics of pooling and lattice operations visible in actual images. Named-city, geographic, or sports-team studies would require suitable image labels and controls for landmarks, text, logos, and backgrounds. The present animal and vehicle labels support the reported comparisons only. Following the distinction between feature hypotheses and circuit evidence in [12], a future causal analysis should test predictions under controlled interventions rather than treating these galleries as recovered computational circuits.



Figure 4: Dataset exemplars and site activation maps for coordinates selected by bird, cat, and dog training contrasts. Each image is followed by its measured 6×6 feature map; intensity uses a shared scale within a row. Titles identify the actual test label, original index, and maximum activation. A selected class does not determine the labels of the highest-activating examples.



Figure 5: The same analysis for horse-, ship-, and truck-associated coordinates. Deer occur among the strongest horse-associated responses, and automobiles among the strongest truck-associated responses. Such overlap motivates inspecting feature combinations rather than assigning a class meaning from a few favorable examples.



Figure 6: Measured class response profiles for all test images, in training RMS units. Selection uses training labels; all plotted means use the separate test images. Broad and overlapping responses remain visible rather than being removed from the gallery.

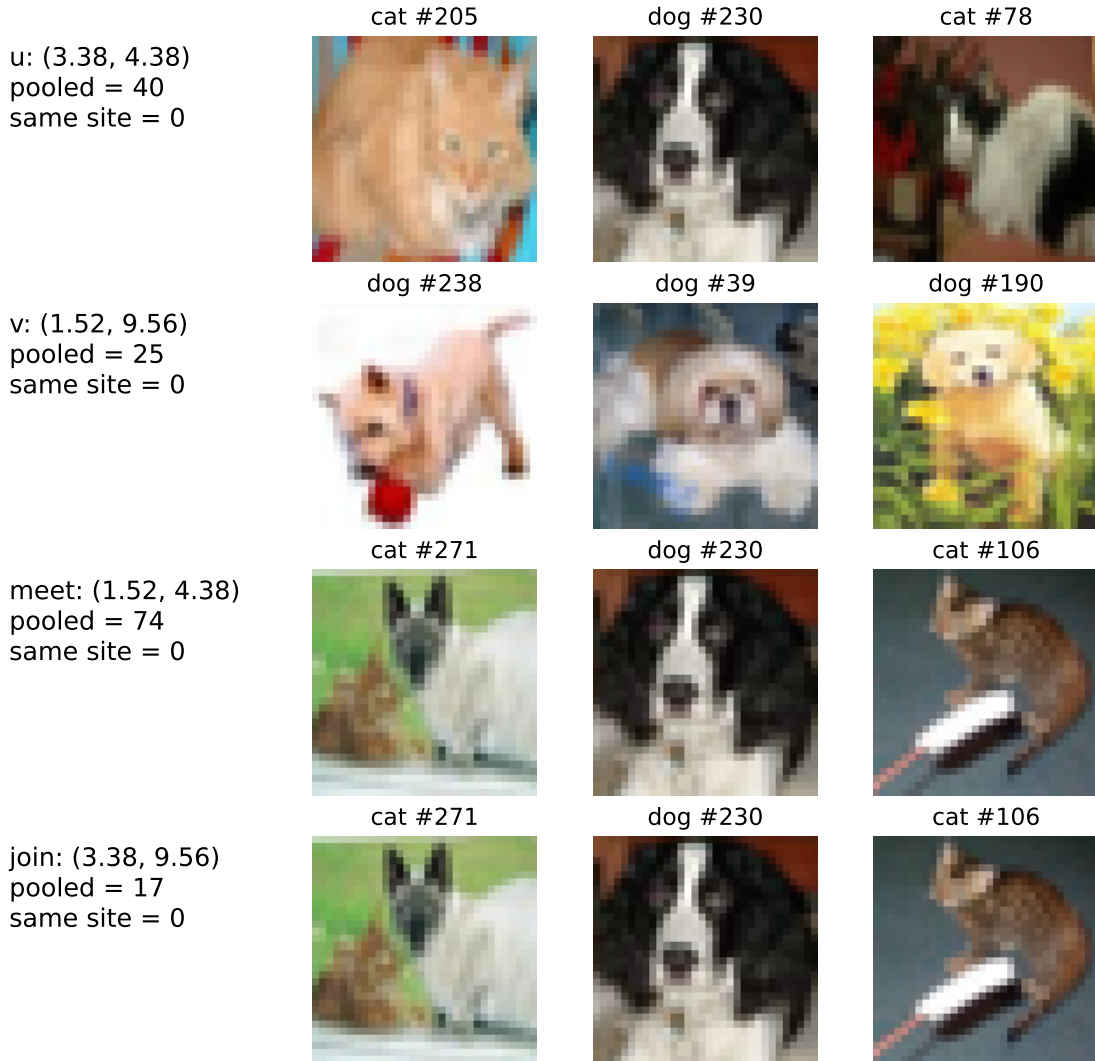


Figure 7: Cat/dog-associated queries, their meet, and their join. Thresholds are displayed in coordinate order (425,150). Counts include all 250 test images. Each row shows up to three matching images ranked by Equation (27); numbers beside the images are original test indices. All four same-site counts are zero.

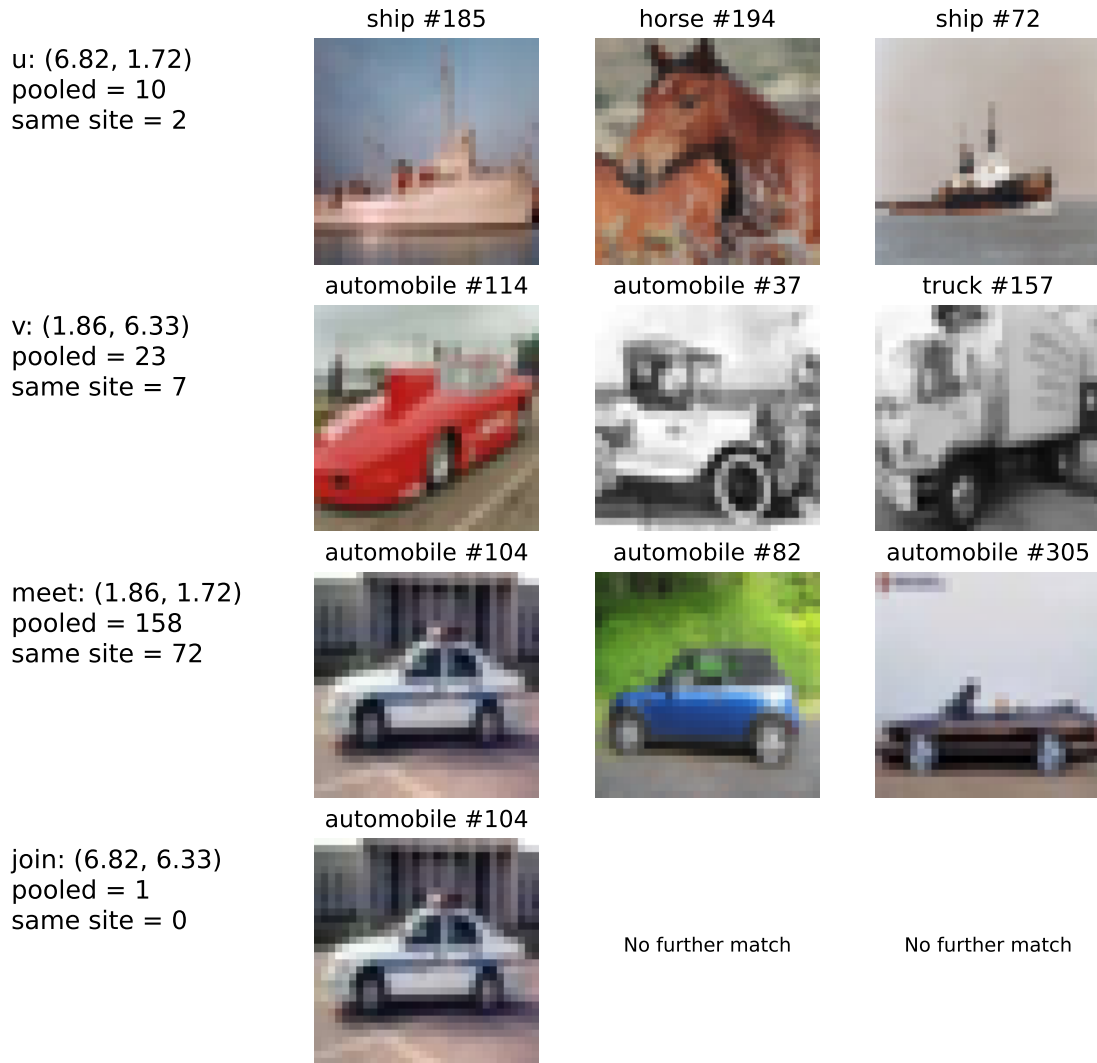


Figure 8: Ship/truck-associated queries on coordinates (78, 461). The meet is broad, while the join has one pooled match, labeled automobile. Its same-site extent is empty. Blank retrieval positions explicitly indicate that no further image satisfies the query.

7 Discussion and Conclusion

Sparse visual codes and order-theoretic descriptions play different roles. The surrogate supplies candidate coordinates; the context states exactly how those coordinates are shared and retrieved. The construction retains the original relation-induced proof pattern and is fully compatible with established pattern structures and ordinally scaled FCA. Its value here is an explicit, checkable specification of visual feature queries rather than a new general representation theorem.

The main vision-specific lesson is that max pooling changes the spatial question. A coordinatewise image query allows separate witnesses for its requirements. A same-site query requires a common witness and generally cannot be represented by a single summary vector. The single-vector criterion and the downset context explain this distinction exactly, while the executed pilot shows that it materially affects the retrieved sets even in a small convolutional network.

The empirical findings also constrain the utility claim. Graded queries outperform the chosen presence baseline, but cosine retrieval is stronger for digit labels. Such labels measure class association, not the correctness of a semantic name attached to a latent coordinate. Replacing all activations by SAE reconstructions checks one aspect of fidelity; it does not establish the causal importance of an individual coordinate or conjunction. Controlled feature interventions, as explored in related vision SAE work [16], address a different question.

Several limitations remain. The digit benchmark uses one small dataset and one fixed split. The natural-image extension uses a deterministic CIFAR-10 subset, one frozen ResNet34, and a surrogate with only moderate reconstruction fidelity. It supplies inspectable examples rather than a second comparative retrieval benchmark. Training seeds and overlapping source draws do not provide independent dataset replications. Training and SAE seeds are varied together, so these runs do not isolate sources of variation. Dead coordinates are common, there is no external spatial annotation, and no vision transformer or transcoder is evaluated. The spatial downset construction is analyzed mathematically but its full lattice is not mined experimentally. These limitations preclude broad claims about vision-model interpretability.

A larger study should fix pretrained backbone and surrogate revisions, verify their activation conventions, and compare at least two vision families on independent natural-image collections. It should include spatially annotated tasks, matched sparsity and reconstruction quality, strong dense and sparse retrieval baselines, and uncertainty estimates whose resampling units match the intended population. Coordinate alignment across independently learned dictionaries requires its own method; coincident index numbers do not supply such alignment. A useful intervention study would compare selected feature edits with random and norm-matched controls while measuring reconstruction error and the backbone’s output change.

The present paper provides a self-contained mathematical specification, an executable digit pilot, measured natural-image feature galleries, and an explicit route to this broader study. It shows how to ask precise questions about shared sparse evidence, and why retaining the spatial meaning of those questions is essential.

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